

Domain Fine-Tuning vs. Retrieval-Augmented Generation for Medical Multiple-Choice Question Answering: A Controlled Comparison at the 4B-Parameter Scale

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Abstract

Practitioners deploying small open-weight large language models (LLMs) for medical question answering face a recurring design choice: invest in a domain-fine-tuned model, or keep a general-purpose model and inject domain knowledge at inference time via retrieval-augmented generation (RAG). We isolate this trade-off by holding model size, prompt template, decoding temperature, retrieval pipeline, and evaluation protocol fixed, and varying only (i) whether the model has been domain-adapted (GEMMA 3 4B vs. MEDGEMMA 4B, both 4-bit quantized and served via Ollama) and (ii) whether retrieved passages from a medical knowledge corpus are inserted into the prompt. We evaluate all four cells of this 2×2 design on the full MedQA-USMLE 4-option test split (1,273 questions) with three repetitions per question (15,276 LLM calls). Domain fine-tuning yields a +6.8 percentage-point gain in majority-vote accuracy over the general 4B baseline (53.3% vs. 46.4%, McNemar $p < 10^{-4}$). RAG over MedMCQA explanations does not produce a statistically significant gain in either model, and in the domain-tuned model the point estimate is slightly negative (−1.9 pp, $p = 0.16$). At this scale and on this benchmark, domain knowledge encoded in weights dominates domain knowledge supplied in context. We release the full experiment code and JSONL traces to support replication.

1 Introduction

Building reliable medical question-answering systems with small, locally-runnable language models is attractive for cost, latency, and privacy reasons, but small models lag larger frontier models on clinical benchmarks [11, 6]. Two practical interventions are available to a practitioner who is constrained to a fixed parameter budget:

1. **Fine-tuning / continued pretraining** on medical corpora, which encodes domain knowledge into the model’s weights (e.g., MEDGEMMA [10] on the GEMMA [3] backbone).
2. **Retrieval-augmented generation (RAG)** [5], which leaves the weights untouched and instead conditions the model on retrieved passages from a domain corpus at inference time.

Both interventions have well-documented benefits on medical benchmarks [13, 12], but they are usually compared in different studies, with different backbones, different evaluation sets, and different retrieval pipelines. The practitioner’s actual question — *at a fixed parameter budget, which intervention buys more accuracy?* — is rarely answered head-to-head.

This paper conducts that head-to-head comparison in a controlled 2×2 design (general vs. domain-tuned model) $\times$ (no RAG vs. RAG) at the 4B-parameter scale. We deliberately use small quantized models served via Ollama because that is the deployment profile most accessible to clinicians, students, and small clinics.

Contributions.

- A controlled, paired comparison of domain fine-tuning and RAG on the full MedQA-USMLE 4-option test split (1,273 questions, 3 repetitions, 15,276 LLM calls).
- Statistical evidence (McNemar’s test on paired predictions) that, at 4B parameters, domain fine-tuning gives a significant accuracy gain whereas a competitive RAG pipeline does not.
- A fully reproducible open-source pipeline (Ollama + ChromaDB + `nomic-embed-text` [7, 1, 8]) with deterministic temperature and per-question repetitions.

2 Related Work

Medical LLMs. Singhal et al. established that scaling plus instruction tuning yields strong USMLE-style performance [11], and GPT-4 demonstrated near- or above-passing performance on USMLE-style items [6]. MEDGEMMA continues this trend in the open-weight regime by adapting GEMMA 3 to medical text and images [10].

Retrieval-augmented generation for medicine. MIRAGE [13] and concurrent work [12] show RAG can lift accuracy on medical QA, especially for general-purpose backbones. The reported gains are typically larger when the retrieval corpus closely matches the question source (e.g., textbook-grounded questions retrieved over the same textbooks).

Direct comparison. Few studies hold the backbone, prompt, and evaluation set fixed while toggling *both* fine-tuning and RAG. Our work closes that gap at a small, deployment-relevant scale.

3 Method

3.1 Design

We use a fully crossed 2×2 design over two binary factors: DOMAIN-TUNED (yes / no) and RAG (yes / no). All other factors — prompt template, decoding temperature, max retrieval budget, evaluation set, repetitions, and aggregation — are held constant. The four resulting setups are summarized in Table 1.

Setup	Backbone	Context
GEMMA3-4B	Gemma 3 4B (general)	question only
GEMMA3-4B+RAG	Gemma 3 4B (general)	question + retrieved passages
MEDGEMMA-4B	MedGemma 4B (domain)	question only
MEDGEMMA-4B+RAG	MedGemma 4B (domain)	question + retrieved passages

Table 1: The four experimental setups. Both backbones are served as 4-bit-quantized GGUF builds via Ollama at temperature 0.1.

3.2 Models

Both backbones are 4B-parameter, 4-bit-quantized, instruction-tuned checkpoints served locally via Ollama [8]. The general backbone is `gemma3:4b` [3]; the domain backbone is a community 4-bit GGUF build of MEDGEMMA 4B [10] (`edwardlo12/medgemma-4b-it-q4_k.m`). Sampling temperature is fixed to 0.1 for all calls.

3.3 Evaluation set

We use the test split of MedQA-USMLE 4-option [4] (1,273 items). Each item is a USMLE Step-style multiple-choice question with four options and a single gold answer.

3.4 Retrieval pipeline

To avoid trivially leaking answers, the retrieval corpus is the *explanation* field of the MedMCQA training split [9] — the questions and answer options of MedMCQA are never indexed, and MedMCQA is disjoint from MedQA. This makes the corpus a textbook-style proxy rather than a leakage path.

The corpus is chunked at 512 tokens with 50-token overlap, embedded with `nomic-embed-text` [7], and stored in a ChromaDB persistent collection [1]. At inference time we form a query from the question text concatenated with all four answer options, retrieve dense candidates, drop chunks with cosine distance above 0.3, and re-rank them with a hybrid score $\alpha \cdot \text{dense} + (1 - \alpha) \cdot \text{lexical}$ ($\alpha = 0.6$). The top $k = 3$ chunks are inserted into the RAG prompt. Across all RAG calls, 96.35% returned a non-empty context (post distance cutoff) with a mean of 2.89 retrieved chunks per call.

3.5 Prompting

Prompts are paraphrases of one another and both enforce the same machine-readable answer contract `{"answer": "A|B|C|D"}`. The RAG prompt prepends the retrieved chunks and explicitly tells the model that the passages *may or may not* be relevant. Both prompts use the same medical-expert persona. The exact strings are in the released code (`src/llm/prompts.py`).

3.6 Repetitions and aggregation

Each (question, setup) pair is evaluated three times. The reported accuracy is the *majority-vote* answer across the three repetitions; ties are broken by the first repetition. We also report *consistency* (the average within-setup agreement across the three repetitions), which is high in all conditions (≥ 0.99 , Table 4), so the choice of aggregation rule has negligible impact on the conclusions.

3.7 Statistical testing

To compare any two setups we use McNemar’s exact paired test [2] on majority-vote correctness over the same 1,273 items. We report the discordant cell counts (*a*-only-correct, *b*-only-correct), the test statistic, the *p*-value, and the binomial 95% confidence interval on accuracy.

4 Results

4.1 Headline accuracy

Majority-vote accuracy and Wilson 95% confidence intervals for the four setups appear in Table 2 and Figure 1. The domain-tuned backbone (MEDGEMMA 4B) leads both with and without

RAG, and the best non-domain setup (GEMMA 3 4B + RAG) does not catch up to the worst domain setup (MEDGEMMA 4B + RAG).

Setup	Accuracy	95% CI	Correct	N
GEMMA3-4B	0.4643	[0.4370, 0.4917]	591	1273
GEMMA3-4B+RAG	0.4729	[0.4456, 0.5004]	602	1273
MEDGEMMA-4B+RAG	0.5137	[0.4863, 0.5411]	654	1273
MEDGEMMA-4B	0.5326	[0.5051, 0.5599]	678	1273

Table 2: Majority-vote accuracy on the MedQA-USMLE 4-option test split.

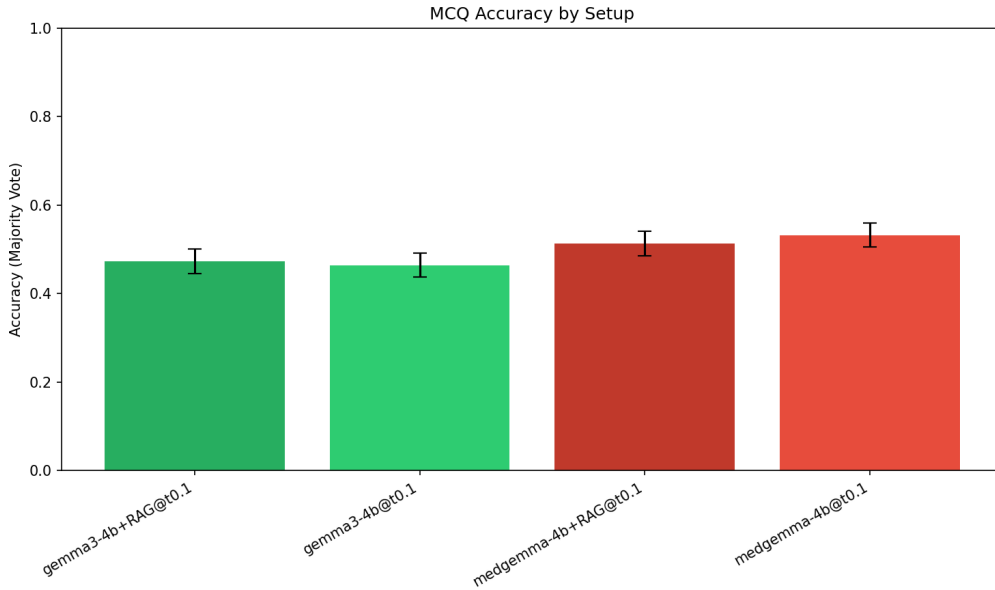


Figure 1: Majority-vote accuracy with 95% confidence intervals. The +6.8 pp gap from domain fine-tuning is significant; the RAG gap is not.

4.2 Pairwise significance

McNemar’s test on paired predictions (Table 3, Figure 2) tells a clear story:

- Every comparison that *crosses* the model boundary (general vs. domain) is significant at $p < 0.005$.
- Neither comparison *within* a model (base vs. +RAG) is significant. For MEDGEMMA 4B the point estimate of the RAG effect is in fact slightly negative (−1.9 pp, $p = 0.16$).

4.3 Consistency

All four setups are highly consistent across repetitions (Table 4); consistency is ≥ 0.99 in every condition. RAG slightly increases consistency on the general backbone and slightly decreases it on the domain backbone, but all four values are within ~ 0.7 pp of each other. The choice of three repetitions therefore mainly serves to expose decoding noise; it is not a major source of variance at $T = 0.1$.

Setup A	Setup B	<i>a</i> -only	<i>b</i> -only	χ^2	<i>p</i> -value
GEMMA3-4B+RAG	GEMMA3-4B	156	145	0.33	0.5644
MEDGEMMA-4B+RAG	MEDGEMMA-4B	124	148	1.94	0.1631
GEMMA3-4B+RAG	MEDGEMMA-4B+RAG	139	191	7.88	0.0050
GEMMA3-4B	MEDGEMMA-4B	144	231	19.72	$< 10^{-4}$
GEMMA3-4B+RAG	MEDGEMMA-4B	179	255	12.96	0.0003
GEMMA3-4B	MEDGEMMA-4B+RAG	176	239	9.26	0.0023

Table 3: McNemar’s test on paired majority-vote correctness. *a*-only / *b*-only are the discordant cells. Comparisons that cross the general/domain backbone boundary are significant; comparisons that toggle RAG within a backbone are not.

Setup	Consistency	Parse-fail rate
GEMMA3-4B	0.9974	0.0000
GEMMA3-4B+RAG	0.9992	0.0000
MEDGEMMA-4B	0.9921	0.0047
MEDGEMMA-4B+RAG	0.9919	0.0039

Table 4: Within-setup agreement across the three repetitions, and fraction of model outputs that did not match the answer schema after two retries.

5 Discussion

The headline finding is straightforward: for a 4B-parameter model answering USMLE-style multiple-choice questions, swapping the general GEMMA 3 backbone for the domain-adapted MEDGEMMA backbone delivers a +6.8 pp accuracy gain that is highly significant on paired data ($p < 10^{-4}$), whereas adding a distance-filtered RAG context retrieved from a textbook-style corpus does *not* produce a significant gain in either backbone.

Why might RAG fail to help here? We see at least four plausible contributors, none of which we can definitively rule out from this experiment alone. *(i)* USMLE items are heavily *reasoning*-driven: the diagnosis or next-best step usually requires integrating findings, not retrieving a fact; even a perfectly relevant passage may not shortcut the chain of reasoning. *(ii)* The corpus (MedMCQA explanations) is broad but not authoritative, and may dilute rather than reinforce the model’s prior. *(iii)* A 4B model has limited capacity to read, ground, and integrate three new chunks while still solving the question; RAG gains in the literature are most clearly demonstrated on larger backbones [13]. *(iv)* MEDGEMMA has likely already absorbed much of the textbook knowledge that the retrieval corpus tries to surface, leaving little marginal information for RAG to add — and possibly some interference when the retrieved passage and the model’s prior conflict.

Practical implication. For a practitioner choosing how to spend an engineering budget on a small, locally-deployable medical QA system, our results suggest that adopting a domain-adapted backbone is a more reliable lever than building a retrieval pipeline. RAG is not actively harmful here, but it is not a substitute for in-weights domain knowledge at this scale.

6 Threats to Validity

Single benchmark. We evaluate only on MedQA-USMLE 4-option. Findings may differ on benchmarks where retrieval is more clearly textbook-grounded (e.g., open-book settings where the answer appears verbatim in the corpus).

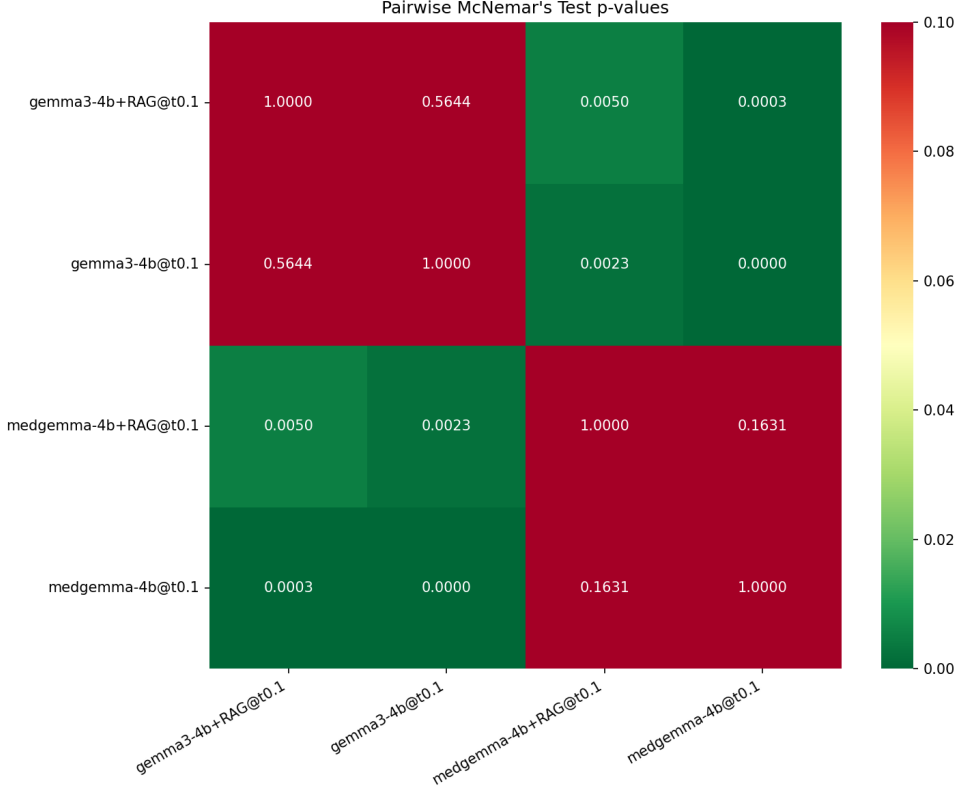


Figure 2: Pairwise McNemar p -values across the four setups. The two non-significant cells are precisely the two RAG-toggle comparisons.

Single corpus and embedding. We use one retrieval corpus (MedMCQA explanations) and one embedding model (`nomic-embed-text`). A different corpus (e.g., UpToDate, clinical guidelines, or MedQA-aligned textbooks) or a stronger embedder could change the RAG result. We mitigated obvious configuration weakness by tuning `max_distance`, `k`, and the hybrid rerank coefficient on a smaller pilot before locking the configuration for the full run.

4-bit quantization. Both backbones are served as 4-bit GGUF builds. Quantization can shift accuracy by a few points and may not affect both backbones equally; the absolute numbers should be read in that light, although the *paired* McNemar comparison still isolates the effect of the manipulated factor.

Single scale. Both models are 4B parameters. We cannot extrapolate to 70B-class or frontier API models, where prior work suggests RAG gains can be larger.

Possible pretraining contamination. MedQA is a public benchmark; either backbone may have seen items during pretraining. We have no reason to believe contamination differentially favors one backbone, but we cannot rule out an absolute-level effect.

7 Conclusion

In a fully paired 2×2 comparison on the full MedQA-USMLE test split, domain fine-tuning of a 4B model significantly outperforms RAG over a textbook-style corpus, while RAG does not significantly help either the general or the domain-tuned backbone. At the deployment-relevant scale of small open-weight models served locally, domain knowledge in the weights appears

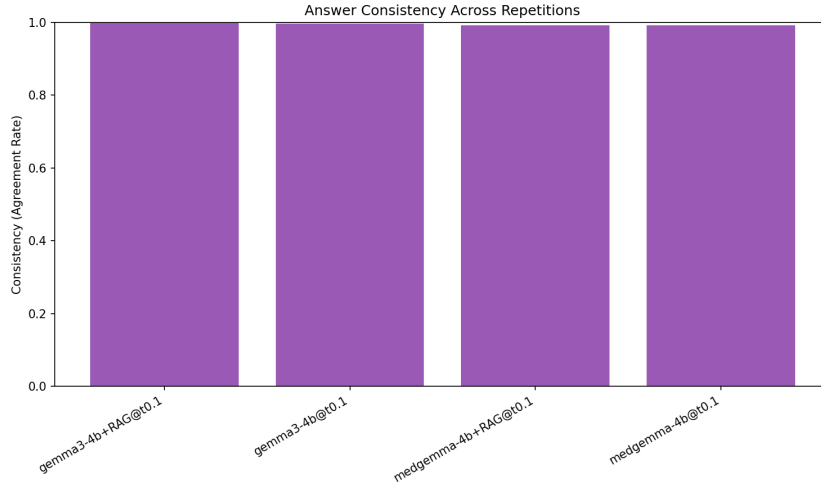


Figure 3: Within-setup consistency across repetitions.

to dominate domain knowledge in the prompt. We hope this controlled comparison helps practitioners prioritize fine-tuning over retrieval pipelines when both options are available.

Reproducibility

Code, configuration files, raw JSONL outputs (15,276 records), and analysis scripts are provided in the project repository. The full experiment can be reproduced end-to-end with:

```
ollama serve
python scripts/build_rag_index.py --config config.yaml
python scripts/run_experiment.py --config config.yaml
python scripts/analyze_results.py --experiment medical_mcq_comparison_full_0.1_3r
```

The exact retrieval, prompt, and decoding hyperparameters are committed in `config.yaml`.

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